

4.5 TF-IDF

Down-weight words that appear in almost every document.

1. Not every word should count the same

One-hot, bag of words, and bag of n-grams treat every token as equally important. *the* and *pneumonia* get the same kind of count. That is a problem. A word that appears in almost every document does not tell you which document you are looking at.

TF-IDF (term frequency–inverse document frequency) reweights the counts. The claim: if word (w) is common in document (d_i) and rare in the rest of the corpus, then (w) is **about** (d_i).

Two pieces multiply:

$$[\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)]$$

Same toy documents as the rest of Week 4. The encoding is still a ($|V|$)-vector per document. The numbers are no longer raw counts.

2. Term frequency

TF asks how often term (t) appears **inside** one document. Longer documents have more room to repeat a word, so we normalize by document length:

$$[\text{TF}(t, d) = \frac{\text{count of } t \text{ in } d}{\text{number of tokens in } d}]$$

In (D_1) = “dog bites man”, each of the three words has TF (= 1/3).

Other textbooks use raw count, ($\log(1 + \text{count})$), or a Boolean. The idea is the same: more mentions inside (d) → larger TF.

3. Inverse document frequency

IDF asks how special the term is **across the corpus**. Stop words (*is*, *are*, *the*) have high TF everywhere and should be down-weighted. Rare content words should be up-weighted.

$$[\text{IDF}(t) = \ln \frac{N}{\text{number of documents that contain } t}]$$

(N) is the number of documents. A word that appears in every document has IDF (= $\ln 1 = 0$) (or a tiny number if the implementation adds smoothing). A word that appears in one document of four has IDF (= $\ln 4$).

Word	Docs that contain it	IDF ($\ln(N/\text{df})$)
dog	(D_1, D_2, D_3)	($\ln(4/3)$ approx 0.29)
bites	(D_1, D_2)	($\ln(4/2)$ approx 0.69)
man	(D_1, D_2, D_4)	($\ln(4/3)$ approx 0.29)
eats	(D_3, D_4)	($\ln(4/2)$ approx 0.69)
meat	(D_3)	($\ln(4/1)$ approx 1.39)
food	(D_4)	($\ln(4/1)$ approx 1.39)

meat and *food* are the most informative unigrams in this toy set. *dog* and *man* are closer to stop words.

4. The product, and sklearn

For (D_3) = "dog eats meat", TF of each word is (1/3). The TF-IDF entries (in vocabulary order dog, bites, man, eats, meat, food) are about

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[0.10, 0, 0, 0.23, 0.46, 0]
```

meat dominates. That is the scheme working.

Your hand calculation will not match `TfidfVectorizer`. sklearn uses a smoothed IDF, sometimes L2-normalizes the rows, and may use ($\log(N/\text{df}) + 1$). Read the docstring before you grade a lab against a spreadsheet. Several published variants exist. The product TF ($\times$) IDF is the shared idea.

5. What still fails

TF-IDF is still a **discrete** representation. *movie* and *film* do not share weight. There is no geometry that says they are related.

The vectors are still **sparse and high-dimensional**. Larger vocabulary $\rightarrow$ more zeros $\rightarrow$ harder learning and more RAM.

OOV is unchanged. A new word has no IDF because it has no document frequency in the training corpus.

Use TF-IDF as the default sparse baseline: `TfidfVectorizer` + linear SVM or logistic regression. Then ask whether a dense embedding (Week 5) is worth the complexity.

6. Practice

1. Compute TF-IDF for *bites* in (D_1). Use the formulas in sections 2 and 3.
2. Why does IDF push *the* toward zero on a news corpus even if *the* has a huge TF?

3. Name two reasons a TF-IDF vector for "the film was long" will not be close to one for "the movie was long."